ML09 Mean, median, range and proportions

Week 3, Tables and statistics | Session 9 of 24 | Machine Learning | New library: numpy

Today's question. What is a typical number of units in a sale, and how spread out are the sales?

Objective. By the end: you can calculate the mean, median, range and a proportion, and say which one an extreme value moves most.

> **WHERE THIS IS USED**
> News reports quote the average income or the share of people who did something. One very large value can drag the mean up, while the median stays calm. Which would you quote for what most boda-boda rides on your road cost?

Before this session: ML08 Select, filter and calculate new columns. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
mean	The total shared out equally: add the values and divide by how many.	Sharing a bag of mandazi equally.
median	The middle value once the values are in order.	The pupil standing in the middle of a line sorted by height.
range	The largest value minus the smallest.	The gap between the tallest and the shortest.
percentile	The value below which a given share of the data falls.	The mark that 75 out of 100 pupils did not beat.
proportion	A part divided by the whole; times 100 it becomes a percentage.	12 pupils out of 20 passed.
outlier	A value far from the rest.	One huge order in a week of small sales.

Block A. Typical value and spread

Look at how many units each sale has. Run the cell and read the four numbers.

Predict. Predict whether the mean or the median of the units column is larger. Write your guess on paper first.

Your guess, written before you run it

```
import numpy as np
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
units = shop["units"]
mean_units = units.mean()
```

```
median_units = np.median(units)
spread = units.max() - units.min()
p75 = np.percentile(units, 75)
print(mean_units, median_units, spread, p75)
```

Run it, then compare. The cell prints about 4.62, 4.5, 7 and 7.0. The mean is the total of 277 divided by 60 sales. The median 4.5 sits between the 30th and 31st values in order. The range is 8 minus 1. The 75th percentile 7 means three quarters of the sales have 7 units or fewer. NumPy is the library behind pandas for numbers, and np is its short name.

Investigate.

1. Why is the median 4.5 and not a whole number?
2. What does the 75th percentile tell a shopkeeper?

Modify. Find the 25th percentile of units.

Block B. One extreme value and a proportion

Add one very large value to a small list and see what moves. Then count a proportion.

Predict. Predict which moves more when 100 joins the list: the mean or the median. Write your guess on paper first.

Your guess, written before you run it

```
small = pd.Series([10, 12, 14, 16])
bigger = pd.Series([10, 12, 14, 16, 100])
print(small.mean(), small.median(), bigger.mean(), bigger.median())
share_big = (units >= 5).sum() / len(units)
print(share_big, share_big * 100)
```

Run it, then compare. The cell prints 13.0, 13.0, 30.4 and 14.0, then 0.5 and 50.0. The outlier 100 pulls the mean from 13 to 30.4 and moves the median only to 14. Half of all sales have 5 units or more: 30 out of 60 is the proportion 0.5, which is 50%.

Investigate.

1. Which summary should you quote when one huge sale sits in the data?
2. Why must you name the whole when you give a proportion?

Modify. Replace 100 with 1000 in bigger and print the mean and the median.

Make

Compare 12 successes out of 20 attempts with 15 out of 50. Calculate each percentage and explain why the larger count does not have to mean the better rate.

Save your work as ML09. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
NameError: name 'np' is not defined	numpy was not imported in this notebook session.	Run import numpy as np first.
The proportion is above 1	You divided the whole by the part.	Divide the part by the whole.

Exit ticket

The mean follows an extreme value, the median stays put, and a proportion needs its whole named.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.

After this session: foundation lab F03 on the fourth day of the week.